This is a sample of writing a paper with Quarto and a journal template

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Abstract

Write something useful here

1 Run some code

2 Make a plot

3 Citations

Here we use a citation to Li et al. (2024), or also (Li et al. 2024).

Link to figure

4 Mathematics uses latex

Proposition 4.1. *Let $x, y \in \Psi$ be statistical graphics. Let $g : \Psi \to \mathbb{R}$ be the Euclidean distance between the data of x and y before the aesthetic function, A is applied, such that $g(x, y) = d(A^{-1}(x), A^{-1}(y))$. Let $h : \Psi \to \mathbb{R}$ be a piecewise distance function that measures the human perception of statistical graphics, defined as:*

$$h(x, y) = \begin{cases} g(x, y) & g(x, y) \geq \epsilon \\ 0 & g(x, y) < \epsilon \end{cases}$$

where ϵ is the JND of our human observer (with $\epsilon > 0$). h is not a metric.

Proof. Assume (Ψ, h) is a metric space. Let $a, b, c \in \Psi$ be three different visualisations, where b is exactly halfway between a and c, such that $g(a, c) = \epsilon$, $g(a, b) = \frac{1}{2}\epsilon$, and $g(b, c) = \frac{1}{2}\epsilon$. Therefore, $h(a, c) = \epsilon$, $h(a, b) = 0$, and $h(b, c) = 0$. As (Ψ, h) is a metric space, the triangular inequality should hold, and we can state

$$h(a, c) \leq h(a, b) + h(b, c)$$

which implies

$$0 \leq \epsilon$$

Therefore, by contradiction, h is not a metric. □

References

Li, W., Cook, D., Tanaka, E. & VanderPlas, S. (2024), 'A Plot is Worth a Thousand Tests: Assessing Residual Diagnostics with the Lineup Protocol', *Journal of Computational and Graphical Statistics* pp. 1–19.